Sheet chapter 3

1- By rocking a boat produces surface water waves on a previously quite lake. He observes that the boat performs 12 oscillations in 20 s; each oscillation producing a wave's crests 15 cm above undisturbed surface of the lake. He further observes that a given waves crest reaches shore, 12 m away in 6 s. What are (a) the period, (b) the speed, (c) the wavelength, and (d) the amplitude of this wave?

2- If the equation of a transverse wave traveling in a string is given by:

$$\mathbf{y(x,t) = (2\ mm)\ sin\ (20\ x - 600\ t)}$$

i- Find the amplitude, frequency, velocity, and wavelength of the wave.

ii- Find the maximum transverse speed of a particle in the string.

3- A uniform cord has a mass of 0.3 kg and a length of 6 m. The cord passes over a pulley and supports a 2 kg object. Find the speed of a pulse traveling along this cord.

4- What is the speed of a traveling wave in a rope of length 2 m and mass 60 g under a tension of 500 N?

5- A piano string having a mass per unit length 5×10^{-3} kg/m is under a tension of 1350 N. Find the speed with which a wave travels on this string.

6- The speed of a wave on a string is 170 m/s when the tension is 120 N. To what value must the tension be increased to raise the wave speed 180 m/s?

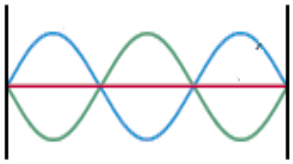
7- A string has linear density 525 g/m and is under tension 45 N. We send a sinusoidal wave with frequency 120 Hz and amplitude 8.5 mm along the string. At what average rate does the wave transport energy?

8- A string 2.7 m long has a mass of 260 g. The tension in the string is 36 N. What must be the frequency of traveling waves of amplitude 7.7 mm to that the average transmitted power is 85 W?

9- Power (P_1) is transmitted by a wave of frequency (f_1) on a string with tension ($\mathcal{T}_1$). What is the transmitted power (P_2) in terms of (P_1): (a) if, instead, the tension of the string is increased to ($\mathcal{T}_2$) = $4\,\mathcal{T}_1$, and (b) if the frequency is decreased to ($f_2 = f_1/2$)?

10- A string fixed at both ends is 8.4 m long and has a mass of 0.12 Kg, it is subjected to a tension of 96 N and starts to oscillate. (a) What is the speed of the waves on the string? (b) What is the longest possible wavelength for the standing waves? (c) Give the frequency of the wave waves on the string.

11- A nylon guitar string has a linear density of 7.2 g/m and is under a tension of 150 N. The fixed supports are 90 cm apart. The string is oscillating in the standing wave pattern as in the corresponding figure. Calculate the (a) speed, (b) wavelength, and frequency of the waves whose superposition gives this standing wave.



12- A cello A-string vibrates in its first normal mode with a frequency of 220 Hz. The vibrating segment is 70 cm long and has a mass of 1.2 g. Find the tension in the string.

13- A taut rope has a mass of 0.18 kg and a length of 3.6 m. What power must be supplied to the rope to generate sinusoidal waves having an amplitude of 0.1 m and a wavelength of 0.5 m and traveling with a speed of 30 m/s?

14- 1.5 cm wire has a mass of 8.7 g and is held under a tension of 120 N. The wire is held rigidly at both ends and set into vibration. Calculate:

i- The velocity of waves on the wire.

ii- The wavelengths and frequencies of the waves produce one or two loops of standing waves on the string.

15- Two identical sinusoidal waves, moving in the same direction along a stretched string, interfere with each other. The amplitude of each wave is 9.8 mm, and the phase difference φ between them is 100°. (a) What is the amplitude of the resultant wave due to the interference, and what is the type of this interference?